Truncation Error of the Multinomial Inversion Series

Uniform smoothing, Wiener's theorem, and why the chi-squared statistic fails

Abstract

The series this codebase sums to obtain an exact multinomial p-value is a trapezoid-rule inverse Laplace transform, and the factor it calls `sinc` is exactly the transform of a uniform smoothing variable $U \sim \mathrm{Unif}[0, W]$: the computation recovers the *density* of $S + U$ at zero, which equals $P(S \leq 0)$ up to the constant W. Reading it that way makes the truncation error tractable. The error is $\sum_{|n|>L} c_n M_S(\gamma + inT)$ with c_n explicit and data-independent, so everything hinges on the decay of the statistic's own transform M_S; the $1/n$ floor that limits it comes from the *edges of the smoothing box*, not from any discontinuity in the statistic. Wiener's theorem then makes precise the observation that a large, diffuse support converges far better than a worst-case analysis predicts: the root-mean-square of M_S over frequency is $M_S(\gamma)/\sqrt{N_{\text{eff}}}$, where N_{eff} is the participation ratio of the tilted measure's atoms. This yields an error estimate computable from terms already summed, which we validate against exact enumeration (agreeing within a factor of 0.14–4.6 across 24 cases). The same diagnostic explains, quantitatively, why the chi-squared statistic ($\lambda = 1$) fails in practice while the log-likelihood-ratio and probability-mass statistics do not: chi-squared's rewards are quadratic in the counts with rationally related coefficients, putting the attainable values on a lattice, and its measured N_{eff} *degrades* towards 1 as K grows ($5.2 \to 1.08$ for $K = 5 \to 20$) while the others improve by five orders of magnitude. At $K \geq 10$ it returns negative p-values.

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1 What the series actually computes

Let $S = \sum_k a_k(X_k)$ be the centred statistic (a_k = `matrix[].reward`), so that the p-value is $p = P(S \le 0)$, and let γ be the exponential tilt chosen by `fill_gamma`. Write $z_n = \gamma + inT$ and $M_S(z) = \mathbb{E}[e^{-zS}]$.

Throughout, L denotes the number of *series terms* retained (the truncation index, so that $|n| \le L$), while N and K keep their usual meaning as the *instance size* — the number of trials and of categories. The two are unrelated and appear together constantly: the standard demo instance is $N = 1000$, $K = 10$, summed to $L \approx 10^3$ terms. (N_{eff}, introduced in §5, is a third and distinct quantity: the effective number of atoms of the tilted measure.)

The factor the evaluators multiply into every term,

$$\mathsf{sinc}(z) \;=\; \frac{1 - e^{-zW}}{zW}, \qquad W = \pi/T,$$

is not an ad-hoc regulariser. It is exactly the two-sided Laplace transform of a uniform random variable:

$$\mathbb{E}\big[e^{-zU}\big] \;=\; \frac{1}{W}\int_0^W e^{-zu}\,du \;=\; \frac{1 - e^{-zW}}{zW}, \qquad U \sim \mathrm{Unif}[0, W],\;\; U \perp S.$$

Hence each stored term is `res_arr`$[n] = M_S(z_n)\,\mathsf{sinc}(z_n) = M_{S+U}(z_n)$, the transform of the *smoothed* variable $Y = S + U$. Convolving with U is what turns the atomic S into something possessing an honest density: each point mass is spread rightwards over a box of width W.

Proposition 1 (The identity the code exploits)**.** *If $-W \le s_{\min}$ and $s_{\max} \le W$, then*

$$p \;=\; P(S \le 0) \;=\; W \cdot f_{S+U}(0).$$

Proof. $f_{S+U}(0) = \sum_j P(S = s_j)\, f_U(-s_j) = \frac{1}{W}\sum_{j:\,-W \le s_j \le 0} P(S = s_j) = \frac{1}{W} P(-W \le S \le 0)$, and $S \ge s_{\min} \ge -W$ makes the truncation vacuous. □

So the computation is density estimation, not CDF evaluation, and the two differ only by the constant W. Inverting by the trapezoid rule on the Bromwich contour,

$$f_Y(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} M_Y(\gamma + it)\,dt \;\approx\; \frac{T}{2\pi}\sum_{n \in \mathbb{Z}} M_Y(z_n),$$

and since $WT = \pi$, Proposition 1 gives $p = \frac{1}{2}\sum_{n\in\mathbb{Z}} M_Y(z_n)$ — precisely what `series()` accumulates, with `res_arr`$[0]$ halved and the negative n folded in by conjugate symmetry. This is the Fourier-series inversion method of Abate and Whitt (1992), specialised to a lattice-free statistic.

Remark 1 (Why $W = \max(s_{\max}, -s_{\min})$)**.** *The trapezoid rule returns the* periodised *density $\sum_m f_Y(2mW)$, so exactness needs $f_Y(2mW) = 0$ for $m \ne 0$. Since* supp $f_Y \subseteq [s_{\min}, s_{\max} + W]$*, the binding image is $m = 1$, requiring $s_{\max} \le W$; Proposition 1 separately requires $-s_{\min} \le W$. Together these are exactly `fill_interval`'s $W = \max(s_{\max}, -s_{\min})$, and the `undersampling` option is the knob that deliberately violates them.*

2 Deconvolution, and a free correctness check

Because sinc is known in closed form and carries no data dependence,

$$M_S(z_n) \;=\; \texttt{res_arr}[n] \,/\, \mathsf{sinc}(z_n)$$

recovers the statistic's own transform by dividing out the kernel — an exact deconvolution. Every diagnostic below is a statement about this recovered sequence.

The first is free. Since $\nu(ds) := e^{-\gamma s} P_S(ds)$ is a *positive* measure with $M_S(z_n) = \hat{\nu}(n)$ its Fourier–Stieltjes coefficients,

$$|M_S(\gamma + inT)| \;\le\; M_S(\gamma) \qquad \text{for every } n, \tag{1}$$

with equality only at $n = 0$. A violation is impossible mathematically, so it is a pure numerical-bug detector, costing one division and one comparison per term. It is not hypothetical: the endpoint-chord tilt bug diagnosed alongside this work violated (1) by roughly nineteen orders of magnitude at $n = 0$ on the standard N=1000, K=10 instance, and would have been caught by this check on the very first term.

Remark 2 ($M_S(\gamma)$ is not the first term of the series)**.** *Quantities like $M_S(\gamma)/p \approx 61$ appear repeatedly below, and they sit oddly beside the observation that the series' first term is* smaller *than p by one to two orders of magnitude. Both are correct; they are separated by exactly the kernel factor. For the demo instance $\gamma W = \pi\gamma/T \approx 2951$, so*

$$\mathsf{sinc}(\gamma W) = \mathbb{E}\big[e^{-\gamma U}\big] \approx \frac{1}{\gamma W} \approx 3.4 \times 10^{-4},$$

and therefore

$$\underbrace{M_S(\gamma) \;\approx\; 61\,p}_{\textit{transform at zero frequency}} \qquad \textit{but} \qquad \underbrace{\texttt{res_arr}[0] = M_S(\gamma)\,\mathsf{sinc}(\gamma W) \;\approx\; p/48}_{\textit{first series term}},$$

the two differing by the factor $61 \times 3.4 \times 10^{-4} \approx 0.021$. $M_S(\gamma)$ is the total mass of the tilted measure ν, and $M_S(\gamma) \ge p$ is just the Chernoff bound; at the saddlepoint the gap is the usual $\gamma\sigma\sqrt{2\pi} \approx 65$, matching the measured 61 *for $\sigma_S \approx 26.1$ (computed properly, not backed out, in §3). The first* term*, by contrast, carries the smoothing kernel, which is suppressed by $1/(\gamma W)$ because U is spread over a width W far exceeding the tilt scale $1/\gamma$.*

This also explains why the sum needs many terms even though it converges quickly: $p = W f_{S+U}(0)$ is a local *quantity, and the $n = 0$ term knows only the total mass. The coefficients μ_n stay near $M_S(\gamma)$ out to the correlation scale $n \sim 1/(T\sigma) \approx 36$, and roughly that many terms of size $\sim M_S(\gamma)/(2W\gamma)$ accumulate to p — consistent with the observed partial sums, which reach* 99.7% *of the final value by $n = 100$.*

3 The saddlepoint characterization of γ

`fill_gamma` chooses γ by minimizing, over real z,

$$L(z) \;:=\; \log M_S(z) + \log \mathsf{sinc}(z) \;=\; \log M_{S+U}(z),$$

i.e. it minimizes the log-MGF of the *smoothed* variable $S + U$, not of S itself. Since $M_{S+U}(z) = \mathbb{E}[e^{-z(S+U)}]$ is an ordinary two-sided Laplace transform of a variable with an honest density, this is

the textbook large-deviations tilt-selection problem, and its stationary point is exactly the condition that the γ-tilted mean of $S+U$ vanishes:

$$E_\gamma[S+U] \;=\; 0. \tag{2}$$

Because $U \sim \mathrm{Unif}[0,W]$ is independent of S, tilting the sum by $e^{-z(S+U)} = e^{-zS}e^{-zU}$ tilts each factor independently by the same z, so (2) splits as

$$E_\gamma[S] \;=\; -E_\gamma[U], \qquad E_\gamma[U] \;=\; -\frac{d}{d\gamma}\log \mathsf{sinc}(\gamma) \;=\; \frac{1}{\gamma} - \frac{We^{-\gamma W}}{1-e^{-\gamma W}}, \tag{3}$$

which for the $\gamma W \gg 1$ regime this codebase always runs in ($\gamma W \approx 2800$ on the demo instance) collapses to $E_\gamma[U] \approx 1/\gamma$: a heavily tilted $\mathrm{Unif}[0,W]$ is, to exponential accuracy, an $\mathrm{Exp}(\gamma)$ variable, since the truncation at W is never felt.

Remark 3 (This resolves the earlier double-saddlepoint confusion)**.** *A first attempt to estimate σ by summing per-category variances independently, as if each X_k were an unconstrained Poisson-type count, gave a value wrong by a factor of 48: it silently imposed $E_\gamma[S]=0$ for a variable with no $\sum_k X_k = N$ constraint, which is not the problem being solved. The count-tilt θ (`l_rate`/ν in the code, `rates.tex`) plays no role in (3) either — it is a numerical rescaling internal to the FFT-tree's $[u^N]$ coefficient extraction, not a second mathematical saddlepoint variable, so this is a* one*-dimensional problem in γ, not a joint one in (θ,γ). The extraction is already performed, correctly, at every z the evaluator is called at; consequently*

$$L_S(z) \;:=\; \log M_S(z)$$

is by construction the ordinary cumulant generating function of S under the $[u^N]$-conditional law, and its first two derivatives at γ are exactly $-E_\gamma[S]$ and $\mathrm{Var}_\gamma(S)$ — no auxiliary machinery is needed beyond evaluating L_S at $\gamma\pm h$ with the same coefficient-extraction routine `fill_gamma` already calls, and differencing.

Measurement. $L_S(\gamma \pm h)$ was evaluated by calling `eval_gammas_fft_thread` directly at three nearby real z and subtracting off the closed-form $\log \mathsf{sinc}(z)$ term (recovered $\log M_S(z)$ via the same deconvolution as §2), then central-differenced; the second derivative was stable to 4 significant figures by $h = 10^{-3}$. Table 1 compares the codebase's default gamma tolerance against a much tighter one, on the N=1000, K=10, $\lambda = 0$ demo instance.

Table 1: The saddlepoint quantities at two `eps_gamma` tolerances.

	`eps_gamma`$= 10^{-2}$ (default)	`eps_gamma`$= 10^{-6}$
γ	0.9164039	0.9159373
γW	2802.19	2800.76
$E_\gamma[S]$ (measured)	-1.371	-1.0945
$E_\gamma[U]$ (closed form, Eq. 3)	1.092	1.0918
residual $E_\gamma[S]+E_\gamma[U]$	-0.279	-0.00276
implied offset from the exact minimizer	$\approx 4\times10^{-4}$	$\approx 4\times10^{-6}$
$\sigma_S = \sqrt{\mathrm{Var}_\gamma(S)}$	26.23	26.13

The residual shrinks by two orders of magnitude when `eps_gamma` tightens by four, consistent with `fill_gamma`'s bracket search landing within a fraction of its stopping width of the true minimizer and the residual gradient being (offset)$\times\mathrm{Var}_\gamma(S)$. This is the decisive difference from θ: γ

converges, on refinement, to a genuine root of (3) — it is not a free numerical parameter, it is an under-converged solution of a well-posed equation, and the default tolerance is loose only because the series itself does not need γ any more precise than that to converge fast.

3.1 What this buys the defective series

Section 2 noted $M_S(\gamma) \approx 61p$, i.e. γ nearly, but not exactly, minimizes $\log M_S$ alone: by (3) the true S-only saddle sits at $\gamma + \Delta\gamma$ with $\Delta\gamma \approx 4 \times 10^{-6}$ (tight column), utterly negligible next to γ itself. So the ordinary saddlepoint density formula (Daniels, 1954),

$$f_S(0) \;\approx\; \frac{M_S(\gamma)}{\sigma_S\sqrt{2\pi}}, \qquad p_{\text{defect}} := W f_S(0), \tag{4}$$

can legitimately be evaluated at the *same* γ the code already computes, using σ_S from Table 1 rather than an independently re-optimized tilt. It describes the "defective" series obtained by dropping the sinc factor from every term before summing (equivalently, summing $M_S(z_n)$ recovered via the deconvolution of §2 directly) — but, as §3.2 shows, only its *bounded part*: that series does not converge at all.

Both series were run on the demo instance (4000 terms, well past stagnation) at both tolerances in Table 1; the defective ratio is insensitive to `eps_gamma` because it depends on $\Delta\gamma$ only at second order:

$$\left.\frac{p_{\text{defect}}}{p}\right|_{\text{measured}} = 2855.4 \qquad \text{(both tolerances agree to 5 significant figures).}$$

Two predictions can be compared against it. The crude one, $\gamma W \approx 2800.8$, used earlier only because $\text{sinc}(\gamma W) \approx 1/(\gamma W)$, is 1.9% low. The one built from (4) using the properly computed $\sigma_S = 26.13$ and $M_S(\gamma) = 1.9669 \times 10^{-17}$ gives

$$\left.\frac{p_{\text{defect}}}{p}\right|_{\text{predicted}} = \frac{W\, M_S(\gamma)}{\sigma_S\sqrt{2\pi}\, p} \;\approx\; 2863.8,$$

which is 0.3% high — roughly six times tighter than the crude estimate. This is the payoff of computing σ correctly: it does not change p (the series is exact and needs no variance to compute it, as it always has), but it does turn "the defective series looks related to a saddlepoint approximation" into a checked, quantitative statement.

3.2 The defective series is slowly divergent, and its drift is the pmf

The measurement above is a plateau, not a limit. To see why, regularise S with a smooth mean-zero kernel ρ_σ (say $N(0,\sigma^2)$) instead of the uniform U. Since $M_{\rho_\sigma}(z) = e^{\sigma^2 z^2/2} \to 1$ pointwise in z as $\sigma \to 0$, every term of the L-term inversion sum for $f_{S+\rho_\sigma}(0)$ tends to the corresponding defective term, so for *fixed* L,

$$\lim_{\sigma\to 0} \Big[L\text{-term sum for } f_{S+\rho_\sigma}(0)\Big] \;=\; \text{defective}(L).$$

But the exact (untruncated) smoothed density does *not* converge as $\sigma \to 0$: it diverges like $1/\sigma$, with $\sigma f_{S+\rho_\sigma}(0) \to P(S=0)/\sqrt{2\pi}$. Both statements hold, and they are consistent, because $L \to \infty$ and $\sigma \to 0$ do not commute: $M_{\rho_\sigma}(\gamma + it)$ has modulus $e^{\sigma^2(\gamma^2 - t^2)/2}$, which tends to 1 pointwise in t but *not* uniformly — it damps $|t| \gtrsim 1/\sigma$, and that damping is exactly what makes the $L \to \infty$ sum converge. Taking $\sigma \to 0$ first discards it.

What survives is a genuine divergence, and both pieces are available in closed form. Writing $M_S(z) = \sum_j p_j e^{-zs_j}$ over the atoms of S and exchanging the two finite sums,

$$\text{defective}(L) \;=\; \frac{T}{2\pi} \sum_{|k|\le L} \sum_j p_j e^{-(\gamma+ikT)s_j} \;=\; \frac{T}{2\pi} \sum_j p_j e^{-\gamma s_j}\, D_L(Ts_j),$$

where $D_L(\theta) = \sum_{|k|\le L} e^{-ik\theta} = \sin\big((L+\tfrac12)\theta\big)/\sin(\theta/2)$ is the Dirichlet kernel. The atom at the origin has $D_L(0) = 2L+1$; every other atom keeps its kernel factor. Hence, exactly,

$$\text{defective}(L) \;=\; \underbrace{(2L+1)\,\frac{T}{2\pi}\,P(S=0)}_{\text{drift}} \;+\; \underbrace{\frac{T}{2\pi} \sum_{j:\, s_j\neq 0} p_j\, e^{-\gamma s_j}\, \frac{\sin\big((L+\frac12)Ts_j\big)}{\sin(Ts_j/2)}}_{=:\,R(L)}, \tag{5}$$

verified against direct summation to $\sim 10^{-14}$ relative. So $R(L)$ is precisely the L-th Dirichlet partial sum of the tilted measure with the atom at the origin deleted, which is why it tracks a density: for a measure as finely divided as this one it approaches the local tilted density at 0, and that is the quantity (4) estimates. It follows that

$$P(S=0) \;=\; \lim_{L\to\infty} \frac{\text{defective}(L)}{(2L+1)\,T/(2\pi)}. \tag{6}$$

Equation (6) is Wiener's theorem (§5) in its direct Cesàro-mean form rather than the mean-square form used there: the average of the Fourier–Stieltjes coefficients detects the atom. It is the saddle-point formula (4) that describes $R(L)$, not the series as a whole.

Checked on a small atomic instance with an atom of mass 0.25 at the origin ($W = 40$, $T = \pi/40$): the defective value grows by exactly 5.000000 between $L = 200$ and $L = 1000$, matching $1600 \cdot (T/2\pi) \cdot 0.25$ to the last digit, while $R(L)$ stays inside a band of width ~ 0.1; and the ratio in (6) runs 0.2666, 0.2447, 0.2502, 0.2501, 0.25002 for $L = 10^2 \ldots 3.2 \times 10^4$.

Can $R(L)$ be corrected for? Since (4) predicts $R(L)$ to a measured 0.3%, subtracting it should expose the drift sooner, and it does — but only linearly in the accuracy of the correction. The drift becomes visible once it exceeds the residual left after subtraction, so a correction good to relative accuracy ε requires

$$L \;\gtrsim\; \varepsilon \cdot \frac{\text{plateau}}{2(T/2\pi)\,p_0} \;=\; \varepsilon \cdot 1.33 \times 10^{24} \qquad \text{(demo instance)}.$$

correction	ε	L needed
none	1	1.3×10^{24}
saddlepoint (4), as measured	3×10^{-3}	4.0×10^{21}
perfect smooth correction	2.9×10^{-8}	3.9×10^{16}

The last row is the obstruction, and it is not a deficiency of the saddlepoint expansion. $R(L)$ does not converge: it is an almost periodic function of L, each non-zero atom contributing an undamped oscillation $\sin\big((L+\frac12)Ts_j\big)$. On the demo instance the defective partial sum wanders between $9.154022245 \times 10^{-16}$ and $9.154022506 \times 10^{-16}$ over $L = 3000 \ldots 19500$ — a spread of 2.6×10^{-23}, or 2.9×10^{-8} relative. That is real structure, not arithmetic: Kahan summation error here is $\sim 10^{-32}$ absolute, and the drift accumulated across the same range is 1.1×10^{-35}, twelve orders of

magnitude below the observed wander. No L-independent correction can track it, so ε cannot be pushed below $\sim 3 \times 10^{-8}$ however many orders of the Daniels expansion are computed (the higher cumulants are available from the same finite differences of $\log M_S$ that gave σ_S, but they sharpen the smooth trend, not the oscillation). Modelling the oscillation itself requires the locations of the neighbouring atoms — which is the problem one was trying to avoid.

Remark 4 (Why the plateau is what gets measured in practice)**.** *On the demo instance the drift is real but invisible. There* $p_0 = 2.10 \times 10^{-36}$ *and* $T/2\pi = 1.64 \times 10^{-4}$*, so each conjugate pair of terms adds* 6.88×10^{-40}*; over the* 4000 *terms actually summed that accumulates to* 2.8×10^{-36}*, roughly twenty orders of magnitude below the measured defective value* 9.15×10^{-16}*. The drift only overtakes the plateau at*

$$L \;\approx\; \frac{9.15 \times 10^{-16}}{6.88 \times 10^{-40}} \;\approx\; 1.3 \times 10^{24}$$

terms. That figure is not a coincidence: it is the same order as the term count required to resolve an individual atom of S *through a truncated inversion at all, since resolving structure at scale* δ *needs frequencies out to* $\sim 1/\delta$*, i.e.* $L \sim W/\delta$*. The pmf is genuinely recoverable from the defective series via* (6)*, and the cost of reading it off is exactly the cost that made direct atom extraction infeasible.*

Remark 5 (This does not tighten the bounds elsewhere in this document)**.** *The certified enclosure of §9 and the Wiener estimate of §5 are unaffected by having a correct* σ_S*, for two different reasons. The certified width's scaling law,* $\mathrm{width}/p \approx 4\gamma W/(L+1)$*, was derived by chaining* $\Phi(0)/M_S(\gamma) \approx 2W/(\sigma\sqrt{2\pi})$ *with* $M_S(\gamma)/p \approx \gamma\sigma\sqrt{2\pi}$*;* σ *cancels between the two, so the final bound only ever needed* γ *and* W*, both already available at no extra cost. The Wiener estimate, meanwhile, is governed by* N_{eff}*, the participation ratio* $(\sum_j w_j)^2 / \sum_j w_j^2$ *of the tilted measure's atoms — a collision statistic, not a spread statistic — and §7 is a standing counterexample to any hope that* σ *could substitute for it: the chi-squared statistic's tilted measure has unremarkable variance, yet its* N_{eff} *collapses toward* 1 *because the lattice structure concentrates mass on few atoms* within *that spread.* σ*'s value in this document is entirely in §3.1's saddlepoint interpretation, not in any truncation-error bound.*

4 The truncation error

Writing $c_n = \mathsf{sinc}(z_n)/2$, the error after keeping $|n| \le L$ is exactly

$$\mathrm{err}_L \;=\; \sum_{|n|>L} c_n \, M_S(z_n). \tag{7}$$

Two structural facts follow.

The $1/n$ floor is the box's edges. $|c_n| \to 1/(2\pi n)$, because f_U is an indicator and indicators have jumps. This is the "Gibbs" behaviour, but located correctly: it is a property of the smoothing kernel, fixed and known, not of the statistic. It also means $\sum_n |c_n|$ diverges, so (7) converges only conditionally — cancellation in $M_S(z_n)$ is not a bonus, it is the entire mechanism.

γ postpones it. $|c_n|$ is essentially *flat* at $1/(2W\gamma)$ for $nT \lesssim \gamma$ and only enters the $1/n$ regime beyond $n^\star \approx \gamma/T = \gamma W/\pi$. For the standard demo instance $\gamma/T \approx 891$, so the 1000-term default sits exactly at that crossover.

5 Wiener's theorem: why diffuse supports converge

The remaining question is how large $|M_S(z_n)|$ is beyond the truncation. For an atomic measure there is no decay in the worst case, but there is a sharp statement about the average:

Proposition 2 (Wiener; see Katznelson 2004)**.** *For a finite measure μ on the circle,* $\lim_{L\to\infty} \frac{1}{2L+1} \sum_{|n|\le L} |\hat{\mu}(n)|^2 = \sum_x |\mu(\{x\})|^2$.

Remark 6 (This is not Parseval, and cannot be)**.** *Proposition 2 looks like an isometry statement, but Parseval does not apply to it, and the reason is exactly why the proposition is useful here. Parseval is an isometry $L^2(\mathbb{T}) \to \ell^2(\mathbb{Z})$; an atomic measure has no L^2 density, and its transform is not in ℓ^2 at all — for $\mu = \delta_0$ one has $\hat{\mu}(n) \equiv 1$, so $\sum_n |\hat{\mu}(n)|^2 = \infty$. The Fourier–Stieltjes transform maps $\mathcal{M}(\mathbb{T})$ into ℓ^∞, where no isometry is available. Since S has finite support, ν is purely atomic, i.e. precisely the case Parseval cannot reach.*

The actual mechanism is diagonal detection rather than isometry. Writing the average as a double integral against the Dirichlet kernel,

$$\frac{1}{2L+1} \sum_{|n|\le L} |\hat{\mu}(n)|^2 = \iint \frac{D_L(x-y)}{2L+1}\, d\mu(x)\, d\bar{\mu}(y), \qquad D_L(t) = \sum_{|n|\le L} e^{-int},$$

and $D_L(t)/(2L+1)$ is bounded by 1 and tends pointwise to $\mathbf{1}_{\{t\equiv 0\}}$. Dominated convergence against the finite product measure leaves the diagonal, whose product mass is $\sum_x |\mu(\{x\})|^2$. Finiteness of μ is what makes the domination legitimate.

There is *an isometry hiding here, but between different objects: the left-hand side is the squared Besicovitch mean-square norm of the almost-periodic sequence $\hat{\mu}$, and the proposition says that norm equals the ℓ^2 norm of the* atomic part alone*. Absolutely continuous components contribute nothing (Riemann–Lebesgue), and so do singular continuous ones. That is the sharp version of the intuition driving this section: a perfectly diffuse measure would give $N_{\text{eff}} = \infty$ and perfect cancellation, and N_{eff} measures how far a genuinely atomic ν sits from that idealisation.*

Applied to ν, with atom weights $w_j = e^{-\gamma s_j} P(S = s_j)$, this says

$$\operatorname{RMS}_n |M_S(z_n)| \;=\; \sqrt{\textstyle\sum_j w_j^2} \;=\; \frac{M_S(\gamma)}{\sqrt{N_{\text{eff}}}}, \qquad N_{\text{eff}} := \frac{\left(\sum_j w_j\right)^2}{\sum_j w_j^2}.$$

N_{eff} is the participation ratio of the tilted measure — the rigorous form of "a large support carrying small probabilities". Combining with $\sum_{n>L} |c_n|^2$ and a random-phase model for the unseen coefficients gives the estimate implemented in `src/multinomial/error_bound.h`:

$$\frac{|\text{err}_L|}{p} \;\approx\; \frac{\sqrt{2}\ \operatorname{RMS}|M_S|\ \sqrt{\sum_{n>L} |c_n|^2}}{p}. \tag{8}$$

This is an *estimate*, not a bound: it assumes the unseen coefficients keep behaving like the seen ones. Section 7 is what happens when they do not.

Remark 7 (Measure the RMS in the trailing window)**.** *N_{eff} must be estimated from the upper half of the computed range. At small n the coefficients are still in their smooth initial decay, and averaging from $n = 1$ measures that decay instead of the plateau, making the estimated N_{eff} grow spuriously in proportion to the number of terms averaged. This was observed directly during implementation before being corrected.*

6 Validation against exact enumeration

For small instances `exhaustive_bisection` gives p exactly, so (8) can be checked against the true error. Table 2 shows $N = 60$, LLR, 1000 terms, quad-precision evaluator (so that floating-point error is negligible against truncation error).

Table 2: Estimated vs. actual relative truncation error, $N = 60$, $\lambda = 0$, 1000 terms.

K	p (exact)	p (series)	actual rel. err	estimate	ratio
3	2.33287×10^{-1}	2.33520×10^{-1}	1.00×10^{-3}	2.06×10^{-3}	2.06
4	2.80668×10^{-1}	2.80414×10^{-1}	9.06×10^{-4}	4.58×10^{-4}	0.51
5	3.45203×10^{-1}	3.45373×10^{-1}	4.95×10^{-4}	1.25×10^{-4}	0.25
6	4.18568×10^{-1}	4.18537×10^{-1}	7.60×10^{-5}	6.14×10^{-5}	0.81

Across all 24 measured cases ($K = 3..6$, $\lambda \in \{0, 1\}$, 250/500/1000 terms) the ratio estimate/actual lies in [0.14, 4.57]. So (8) is a reliable order-of-magnitude predictor, and it is *not* conservative — it under-predicts in most rows. It should be used to decide how many terms to take, not to certify a result.

7 The chi-squared statistic fails, and the diagnostic says why

The reward functions in `src/rewards.h` are arithmetically very different:

$$\lambda = 0: \; a_k(i) = -i \log \frac{i}{N p_k}, \qquad \lambda = 1: \; a_k(i) = -\tfrac{1}{2}\Big(\frac{i^2}{N p_k} - i\Big).$$

The chi-squared rewards are quadratic in i with rationally related coefficients, so the attainable values of S sit on (or very near) a lattice. A lattice support makes $\hat{\nu}$ periodic: the coefficients *revive* to full size instead of continuing to cancel, and the mechanism of Section 5 collapses. Table 3 measures this directly at $N = 1000$ with irrational $p_k \propto \sqrt{k+1}$.

Three things are worth stating plainly.

1. **Chi-squared returns negative p-values** at $K = 10$ and 15, and a value at $K = 20$ that is nine orders of magnitude above the comparable LLR figure. These are not marginal accuracy losses; the results are meaningless.

2. **Growing the support makes chi-squared worse, not better.** N_{eff} runs $5.21 \to 1.43 \to 1.21 \to 1.08$ as K goes $5 \to 20$: the tilted measure behaves as though it had *one* atom, no matter how many it actually has. Over the same range LLR improves by five orders of magnitude, $1.6 \times 10^5 \to 1.7 \times 10^{10}$. This is the quantitative form of the practical observation that motivated this note.

3. **The revival column identifies the cause without knowing it in advance.** At $\lambda = 1$, $|M_S|$ returns to within 0.4% of its maximum at high frequency; at $\lambda = 0$ it stays below 0.6% of it. The diagnostic is cheap, requires no reference solution, and flags the failure on any instance.

Table 3: $N = 1000$, irrational p_k, 1000 terms. "revival" is $\max_{n>L/4} |M_S(z_n)|/M_S(\gamma)$; a value near 1 means the coefficients never decayed.

K	statistic	p	N_{eff}	revival	est. rel. err
5	LLR (λ=0)	1.36×10^{-15}	1.6×10^{5}	0.018	1.1×10^{-3}
5	chi-sq (λ=1)	-3.45×10^{-11}	5.21	0.835	2.96
5	pmf (λ=NaN)	1.12×10^{-15}	1.5×10^{5}	0.019	1.2×10^{-3}
10	LLR	3.21×10^{-19}	1.4×10^{8}	0.0046	3.3×10^{-5}
10	chi-sq	-9.41×10^{-9}	1.43	0.980	0.74
10	pmf	1.84×10^{-19}	1.8×10^{8}	0.0041	3.0×10^{-5}
15	LLR	1.98×10^{-18}	2.1×10^{9}	0.0051	7.0×10^{-6}
15	chi-sq	-8.32×10^{-7}	1.21	0.982	0.32
15	pmf	7.95×10^{-19}	2.4×10^{9}	0.0049	6.5×10^{-6}
20	LLR	2.02×10^{-19}	1.7×10^{10}	0.0053	2.3×10^{-6}
20	chi-sq	5.16×10^{-6}	1.08	0.996	0.29
20	pmf	5.18×10^{-20}	3.2×10^{10}	0.0046	1.7×10^{-6}

8 What the density view buys

The reframing is not only presentational. Treating the object as a density supplies regularity that is simply invisible when one works with the tilted measure ν, and that regularity is free, a priori, and closed-form.

Proposition 3 (Free regularity)**.** $f_Y(y) = \frac{1}{W} P\big(y - W \le S \le y\big)$ *satisfies*

$$\|f_Y\|_\infty \;\le\; \frac{1}{W}, \qquad \mathrm{TV}(f_Y) \;\le\; \frac{2}{W}.$$

Proof. The first is immediate. For the second, as y increases each atom of S enters the window $[y-W, y]$ exactly once and leaves exactly once, contributing $P(S = s_j)/W$ to the variation on each occasion; summing gives $2/W$. □

Three consequences.

A closed-form bound replaces the LP. ν is a measure with no density and no bounded variation, so bounding $\int R_L \, d\nu$ required certifying diffuseness from the computed moments (Fejér-kernel interval bounds, or the full moment-problem LP). With Proposition 3 the classical Dirichlet–Jordan rate for functions of bounded variation applies directly, and the a priori bound can be written down rather than solved for:

$$\frac{|\mathrm{err}_L|}{p} \;\lesssim\; \frac{C}{L} \cdot \frac{\gamma W}{\pi} \;=\; \frac{C}{L} \cdot \frac{\gamma}{T},$$

the γ/T being the same crossover frequency of Section 4. No optimisation, no certification step, no unverifiable extrapolation.

The $1/n$ floor is the kernel's, and one of its two edges is negotiable. This is the content of Remark 8 below: only the jump at $y = 0$ is forced by the estimand.

The two regimes are unified. Compare the a priori bound above with the Wiener estimate (8). They differ by the cancellation factor $\sqrt{N_{\text{eff}}}$: the closed-form bound is what you get when the coefficients do *not* cancel, and (8) is what you get when they do. This is not an abstraction — it is exactly the $\lambda = 0$ versus $\lambda = 1$ split of Section 7. For chi-squared the measured $N_{\text{eff}} \to 1$, cancellation is absent, and the pessimistic closed-form bound is not pessimistic at all: it is the truth, which is why that statistic returns garbage at term counts where LLR has ten correct digits. For LLR $N_{\text{eff}} \sim 10^{10}$ and the same bound overstates the error by five orders of magnitude.

So the density view does not improve the sharp worst-case order — that remains $\Theta(1/L)$ for any framing, by Section 9 — but it makes the worst case computable in closed form, explains when it is attained, and supplies N_{eff} as the single measured number interpolating between the two.

9 The certified enclosure: LP duality with exact constants

Everything so far has been an estimate. This section gives what can actually be *certified* from the computed data together with the single structural fact $\nu \geq 0$, with explicit constants and proofs.

9.1 The elementary enclosure

Everything Corollary 1 needs is contained in this subsection; the linear programming of §9.4 is context, not a prerequisite.

Write $\mu_n := M_S(z_n)$ for the computed data and $\nu^\star(ds) = e^{-\gamma s} P_S(ds)$ for the true tilted measure, so that $p = \int f \, d\nu^\star$ with $f(s) = e^{\gamma s} \mathbf{1}_{s \leq 0}$.

Proposition 4 (Sandwich enclosure)**.** *Let P and Q be trigonometric polynomials of degree $\leq L$ with $Q \leq f \leq P$ pointwise on $[-W, W]$, and write λ_n^P, λ_n^Q for their coefficients in the e^{-inTs} basis. Then*

$$\sum_{|n| \leq L} \lambda_n^Q \mu_n \;\leq\; p \;\leq\; \sum_{|n| \leq L} \lambda_n^P \mu_n.$$

Proof. $\nu^\star \geq 0$ and $P - f \geq 0$ give $\int f \, d\nu^\star \leq \int P \, d\nu^\star$. Expanding P in its basis and using that the data are by definition $\nu^\star$'s own transform values,

$$\int P \, d\nu^\star = \sum_{|n| \leq L} \lambda_n^P \int e^{-inTs} \, d\nu^\star(s) = \sum_{|n| \leq L} \lambda_n^P \mu_n.$$

The minorant case is identical with the inequality reversed. □

Two things are worth making explicit, because they are the whole reason the enclosure is valid for *every* L rather than only asymptotically. First, positivity of $\nu^\star$ is used exactly once, and it is the only structural input. Second, no approximation is involved: the μ_n are not estimates of the true measure's moments, they *are* the true measure's moments, so $\nu^\star$ automatically satisfies every constraint that the data impose. Truncating at L discards information but never invalidates what remains — which is why enlarging L can only shrink the enclosure, never move it off p.

9.2 An explicit admissible pair

Selberg's construction supplies majorants and minorants of an *interval indicator* with sharp L^1 gap (Vaaler, 1985; Montgomery, 1994): for any arc I of $\mathbb{R}/\mathbb{Z}$ and any $L \geq 0$ there are trigonometric polynomials of degree $\leq L$ with

$$q_I \leq \mathbf{1}_I \leq p_I, \qquad \hat{p}_I(0) - \hat{q}_I(0) = \frac{2}{L+1}, \qquad \big|\hat{p}_I(n) - \hat{\mathbf{1}}_I(n)\big|, \; \big|\hat{q}_I(n) - \hat{\mathbf{1}}_I(n)\big| \leq \frac{1}{L+1}. \tag{9}$$

Our f is not an indicator, and the crude sandwich $0 \le f \le \mathbf{1}_{[-W,0]}$ is far too lossy: it yields an upper bound near $\frac{1}{2}\mu_0$, some thirty times p. The right move is the layer cake.

Proposition 5 (One-sided approximation of the tilted step)**.** *For $f(s) = e^{\gamma s}\mathbf{1}_{s\le 0}$ on the period $[-W, W)$ and any $L \ge 0$ there exist trigonometric polynomials $Q \le f \le P$ of degree $\le L$ whose coefficients in the e^{-inTs} basis satisfy*

$$\lambda_0^P - \lambda_0^Q = \frac{2}{L+1}, \qquad \left|\lambda_n^P - \lambda_n^Q\right| \le \frac{2}{L+1} \quad (1 \le |n| \le L).$$

Proof. Since $0 \le f \le 1$, the layer-cake identity gives $f(s) = \int_0^1 \mathbf{1}_{I_t}(s)\,dt$ with $I_t = \{f > t\} = (\gamma^{-1}\log t,\, 0] \cap [-W, 0]$, which is an arc for every $t \in (0, 1]$. Put $P := \int_0^1 p_{I_t}\,dt$ and $Q := \int_0^1 q_{I_t}\,dt$; both are trigonometric polynomials of degree $\le L$, and integrating $q_{I_t} \le \mathbf{1}_{I_t} \le p_{I_t}$ over t gives $Q \le f \le P$ pointwise. For the coefficients, $\lambda_0^P - \lambda_0^Q = \int_0^1 \big(\hat{p}_{I_t}(0) - \hat{q}_{I_t}(0)\big)dt = 2/(L+1)$ by (9), and for $n \ne 0$ the triangle inequality through $\hat{\mathbf{1}}_{I_t}(n)$ gives $|\lambda_n^P - \lambda_n^Q| \le \int_0^1 2/(L+1)\,dt = 2/(L+1)$. □

Corollary 1 (Certified enclosure)**.** *With P, Q as above,*

$$\sum_{|n|\le L} \lambda_n^Q \mu_n \;\le\; p \;\le\; \sum_{|n|\le L} \lambda_n^P \mu_n, \qquad \text{width} = \sum_{|n|\le L} \big(\lambda_n^P - \lambda_n^Q\big)\mu_n \;\le\; \frac{2}{L+1} \sum_{|n|\le L} |\mu_n|.$$

The $n = 0$ contribution to the width is exactly $2\mu_0/(L+1) = 2M_S(\gamma)/(L+1)$.

9.3 What the certificate is actually worth (measured)

The width of Corollary 1 is a finite computation over data already in hand, because the Selberg pair's difference is explicit. Combining Vaaler's bound $|V_L - \psi| \le F_L/(2L+2)$ with $\mathbf{1}_{[a,b]} = (b-a) + \psi(\cdot - b) - \psi(\cdot - a)$ gives

$$p_I - q_I = \frac{F_L(\cdot - a) + F_L(\cdot - b)}{L+1},$$

F_L being the Fejér kernel of mean 1. Averaging over the layer cake of Proposition 5 (whose arcs are $I_t = (\gamma^{-1}\log t, 0]$, so $b \equiv 0$) yields a closed form for the width in terms of the computed coefficients alone:

$$\text{width} = \frac{1}{L+1}\Big(\Phi(0) + \int_0^1 \Phi\big(a(t)\big)\,dt\Big), \qquad \Phi(c) := \sum_{|n|\le L}\Big(1 - \tfrac{|n|}{L+1}\Big)e^{inTc}\mu_n = \int F_L(s-c)\,d\nu(s), \tag{10}$$

with $a(t) = \max(-W, \gamma^{-1}\log t)$. Note $\Phi \ge 0$ pointwise, since $F_L \ge 0$ and $\nu \ge 0$ — a further free self-check, and one that caught an uninitialised-buffer bug while this was being measured. Table 4 evaluates (10) on the demo instance.

The verdict is unambiguous, and it goes against the optimistic reading: **the true signed width sits at roughly 90 times its $n = 0$ leading term**, i.e. about $10.8\,p$ at $L = 1024$. The certificate does not merely fail to be tight, it fails to establish that $p > 0$. An earlier draft of this note quoted the leading term, $2M_S(\gamma)/((L+1)p) \approx 122/(L+1) \approx 12\%$, as the best case; the best case is not attained, and the crude absolute-value bound of order $12\,p$ was the accurate prediction.

The reason is a scale mismatch that (10) makes explicit. $\Phi(0) = \int F_L\,d\nu \approx 2W\rho_\nu(0)$ is a *local density* quantity, whereas $M_S(\gamma)$ is a *total mass*; their ratio is therefore

$$\frac{\Phi(0)}{M_S(\gamma)} \;\approx\; \frac{2W}{\sqrt{2\pi}\,\sigma} \;=\; 90.4 \qquad \text{against } 90.6 \text{ measured.}$$

Table 4: Measured Selberg width, $N = 1000$, $K = 10$, $\lambda = 0$ (double-double evaluator).

terms	width/p	n=0 term/p	$\Phi(0)/M_S(\gamma)$
256	39.6	0.477	83.4
512	21.0	0.239	88.2
1024	10.8	0.120	90.6
2048	5.45	0.060	91.8
4096	2.74	0.030	92.4

Propagating this through (10) with the saddlepoint gap $M_S(\gamma)/p \approx \gamma\sigma\sqrt{2\pi}$ collapses everything to

$$\frac{\text{width}}{p} \;\approx\; \frac{4\gamma W}{L+1} \;\approx\; \frac{11800}{L+1}, \tag{11}$$

predicting 11.5 at $L = 1024$ against 10.8 measured, with the $1/L$ decay visible in every row.

Read as a requirement, (11) says the certificate needs $L \gtrsim 4\gamma W \approx 1.1 \times 10^4$ merely to be non-vacuous and $L \sim 10^6$ for one percent, while the series it is certifying has converged by $n \sim 1/(T\sigma) \approx 36$. The enclosure therefore costs some three hundred times more terms than the computation it validates. This is not a defect of Selberg's construction — the same $2W/(\sqrt{2\pi}\sigma)$ mismatch afflicts any bound built from the moments alone, and §9.4's LP can only improve the constant, not the $4\gamma W/L$ law. Whether it is worth paying is an application decision; on smaller instances it can be, since at $N = 60$, $K = 5$ the same computation gives a width of 7% at 4096 terms, the tilt gap there being $M_S(\gamma)/p = 3.9$ rather than 61.

9.4 How far from sharp: the moment LP

Corollary 1 fixes *one* admissible pair. Optimising over all of them is a linear program, and its value is the exact information content of L moments. With $R_\lambda(s) = \sum_{|n|\le L} \lambda_n e^{-inTs}$, the sharp enclosure is

$$p^{\pm} \;=\; \begin{matrix}\sup\\ \inf\end{matrix} \left\{ \int f\,d\nu \;:\; \nu \in \mathcal{M}_+([-W,W]),\;\; \int e^{-inTs}\,d\nu(s) = \mu_n,\;\; |n| \le L \right\}, \tag{12}$$

whose duals are exactly the one-sided approximation problems

$$p^+ = \inf_{R_\lambda \ge f} \sum_{|n|\le L} \lambda_n \mu_n, \qquad p^- = \sup_{R_\lambda \le f} \sum_{|n|\le L} \lambda_n \mu_n, \tag{13}$$

weak duality between them being Proposition 4 verbatim. Strong duality holds by standard semi-infinite linear programming: μ_0 fixes the total mass, so the feasible set is weak-$*$ compact, and f is bounded and Borel.

Proposition 6 (Validity and monotonicity in L)**.** *$p \in [p_L^-, p_L^+]$ for every L, and the intervals are nested: $p_L^- \le p_{L+1}^- \le p \le p_{L+1}^+ \le p_L^+$.*

Proof. The true measure $\nu^\star$ is itself feasible in (12) — it is nonnegative, it is supported in $[-W,W]$, and its transform values at $|n| \le L$ are precisely the data μ_n — so $p = \int f\,d\nu^\star$ is the objective evaluated at a feasible point, hence lies between the infimum and the supremum. Raising L to $L+1$ appends two equality constraints, shrinking the feasible set; a supremum over a smaller set cannot increase and an infimum cannot decrease. □

This is worth stating separately because it is the only reason the enclosure is trustworthy at *every* truncation rather than merely in the limit, and the argument is not asymptotic: the μ_n are not estimates of $\nu^\star$'s moments, they are those moments, so no amount of truncation can put $\nu^\star$ outside the feasible set. (In floating point the μ_n carry the evaluator's own error, so the enclosure is certified modulo that — which is exactly what the positivity ceiling (1) is there to police.)

Computing (12) is a semidefinite program, via the following classical parametrisation of the nonnegativity constraint.

Proposition 7 (Riesz–Fejér)**.** *Let $T(\theta) = \sum_{|n|\le L} c_n e^{in\theta}$ be real-valued with $T \ge 0$ for all real θ. Then there is an* algebraic *polynomial $q(z) = \sum_{k=0}^{L} a_k z^k$ in the complex variable z with*

$$T(\theta) \;=\; \bigl|q(e^{i\theta})\bigr|^2 \qquad \text{for all } \theta.$$

The apparent type mismatch — a periodic T equal to a polynomial — dissolves once one notes that q is *not* evaluated at θ. It is evaluated at $z = e^{i\theta}$, i.e. restricted to the unit circle, and $\theta \mapsto q(e^{i\theta})$ is manifestly 2π-periodic. The right way to see the statement is that a trigonometric polynomial of degree L *is* the restriction to $|z| = 1$ of the Laurent polynomial $\sum_{|n|\le L} c_n z^n$; multiplying by z^L clears the poles and leaves an ordinary polynomial of degree $2L$, so trigonometric and algebraic polynomials are two readings of one object, related by $z = e^{i\theta}$. Reality of T forces $c_{-n} = \overline{c_n}$, which makes the roots of that degree-$2L$ polynomial come in pairs $\{z_j,\, 1/\overline{z_j}\}$, and $T \ge 0$ forces every root actually on the circle to have even multiplicity; collecting one root from each pair (all inside the closed disc) and normalising gives q.

For the program, expanding $|q(e^{i\theta})|^2 = \sum_{j,k} a_j \overline{a_k} e^{i(j-k)\theta}$ shows $c_n = \sum_{j-k=n} a_j \overline{a_k}$, which is the statement that the coefficient vector is read off the diagonals of the rank-one positive semidefinite matrix $G = aa^*$. Relaxing "rank one" to "positive semidefinite" costs nothing — a sum of squares is still nonnegative, and Proposition 7 says one square already suffices — so

$$T \ge 0 \text{ on the circle} \iff \exists\, G \succeq 0 \text{ in } \mathbb{C}^{(L+1)\times(L+1)} \;\text{ with }\; c_n = \sum_{j-k=n} G_{jk},$$

an exact linear matrix inequality in the unknowns λ. Since f is not itself a trigonometric polynomial, the constraint $R_\lambda - f \ge 0$ is imposed arcwise: either reduce to indicators by the layer cake above, or discretise s and control the grid error by a Bernstein inequality on R_λ.

9.5 Using an enclosure

Given any certified $[p^-, p^+]$, the series value $P_L = \sum_{|n|\le L} c_n \mu_n$ acquires a rigorous error bar,

$$|p - P_L| \;\le\; \max\bigl(p^+ - P_L,\; P_L - p^-\bigr),$$

with no appeal to the random-phase model of Section 5.

9.6 What remains uncertifiable

Estimate (8) is valid exactly when the unseen coefficients keep cancelling, and the revival statistic of Section 7 is a direct, cheap test of that premise on the computed range. Extending it beyond L would need quantitative non-resonance — a Diophantine statement about the reward values, plausible for the logarithmic ($\lambda = 0$) rewards and provably false for the lattice-valued ($\lambda = 1$) ones. The moment-problem enclosure above is the part that needs no such hypothesis.

Remark 8 (The far edge of the kernel is *not* free)**.** *Proposition 1 is a statement about f_U at a single point, but the trapezoid rule returns the periodised density $\sum_m f_Y(2mW)$, so what the constraint really binds is the periodised kernel $f_U^{\mathrm{per}}(y) = \sum_m f_U(y+2mW)$. Correctly stated, exactness requires*

$$f_U^{\mathrm{per}}(y) = \tfrac{1}{W} \quad \text{on } [0, -s_{\min}], \qquad f_U^{\mathrm{per}}(y) = 0 \quad \text{on } [-s_{\max}, 0),$$

and those two arcs occupy $s_{\max} - s_{\min}$ of the period $2W$. The room left over is therefore $2W - (s_{\max} - s_{\min})$, which vanishes *in the symmetric case $s_{\max} = -s_{\min} = W$ — the usual one. Extending the box to the right by $B > 0$ pushes* $\operatorname{supp} f_Y$ *out to $s_{\max} - s_{\min} + B > 2W$, and the image at $y = 2W$ folds back onto $y = 0$, corrupting exactly the value being computed. So the earlier suggestion that the far edge could be smoothed for free was wrong: it is free only to the extent the support is asymmetric, and enlarging W to make room shrinks $T = \pi/W$ and costs truncation accuracy directly.*

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